RAJALAKSHMI ENGINEERING COLLEGE

Aashish Durai M 251801002-2025-AIDS

A2

Started on	Saturday, 8 August 2026, 1:59 PM
State	Finished
Completed on	Saturday, 8 August 2026, 2:12 PM
Time taken	13 mins 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int n;
    int i=1;
    int s=1;
    int count=2;
    scanf("%d",&n);
    while(s<=n){
        count++;
        i++;
        count++;
        s+=i;
        count++;

    }
    count++;
    printf("%d",count);
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Aashish Durai M 251801002-2025-AIDS

A2

Started on	Saturday, 8 August 2026, 2:14 PM
State	Finished
Completed on	Saturday, 8 August 2026, 2:37 PM
Time taken	23 mins 45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    if(n==1){
        c++;
        c++;
        //printf("*");
    }else{
        c++;
        for(int i=1;i<=n;i++){
            c++;
            for(int j=1;j<=n;j++){
                c++;
                c++;
                c++;
                //printf("*");
                //printf("*");
                break;
            }
            c++;
        }
    c++;
    printf("%d",c);
    }
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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A2

Started on	Monday, 24 August 2026, 2:00 PM
State	Finished
Completed on	Monday, 24 August 2026, 2:08 PM
Time taken	7 mins 24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int num, i;
    int count = 0;

    scanf("%d", &num);

    for (i = 1; i <= num; ++i)
    {
        count++;

        if (num % i == 0)
        {
            count++;
        }

        count++;
    }

    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Aashish Durai M 251801002-2025-AIDS

A2

Started on	Monday, 24 August 2026, 2:08 PM
State	Finished
Completed on	Monday, 24 August 2026, 2:09 PM
Time taken	1 min 40 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int c = 0;

    scanf("%d", &n);

    c++;

    for (int i = n / 2; i < n; i++)
    {
        c++;

        for (int j = 1; j < n; j = 2 * j)
        {
            c++;

            for (int k = 1; k < n; k = k * 2)
            {
                c++;
                c++;
            }

            c++;
        }

        c++;
    }

    c++;

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Aashish Durai M 251801002-2025-AIDS

A2

Started on	Monday, 24 August 2026, 2:10 PM
State	Finished
Completed on	Monday, 24 August 2026, 2:12 PM
Time taken	2 mins 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, rev = 0, remainder;
    int count = 0;

    scanf("%d", &n);

    count++;

    while (n != 0)
    {
        count++;
        remainder = n % 10;
        count++;
        rev = rev * 10 + remainder;
        count++;
        n /= 10;
        count++;
    }

    count++;
    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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